

Problem 3

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Problem 1 The velocity of the object moving along a straight line is given by $v(t) = t - 5$, $0 \leq t \leq 10$, where t is in seconds and v in ft/s.

- (a) Use geometry to evaluate the displacement of the object on the time interval $[0, 10]$. Sketch the graph of the velocity and shade the relevant region.
[3]
- (b) Compute the displacement on the interval $[0, 10]$ using antiderivatives of the function v .
- (c) Use geometry to evaluate the displacement of the object on the time interval $[0, 8]$. Sketch the graph of the velocity and shade the relevant region.
[3]
- (d) Compute the displacement on the interval $[0, 8]$ using antiderivatives of the function v .